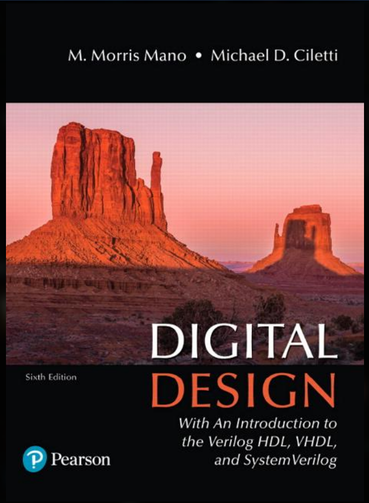


Chapter 1
Digital Systems and Binary Numbers

LEC04/LAB04 UPLOADED

CONVERSION

From Base-r to Base-r′

ARITHMETIC

ADDITION

PADDING

+	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
				.			

					1		
+	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
				.		0	4

$$\frac{16}{16} = 1\ r\ 0$$

		1	1		1		
+	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
			A	.	1	0	4

MULTIPLY

At Home

SUBTRACTION

SUBTRACTION

Example I, Base-10

PADDING

−	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
				.			

IS IT POSSIBLE?

–	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
				.			

			BORROW		-1 →	+10	
−	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
				.		= 10+5-6 = 9	4

		+10					
	-1	-1	+10		-1	+10	
−	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
		8	9	.	0	9	4

		+10					
	-1	-1	+10		-1	+10	
−	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
		8	9	.	0	9	4

	+10	+10					
-1	-1	-1	+10		-1	+10	
−	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
	5	8	9	.	0	9	4

	+10	+10					
-1	-1	-1	+10		-1	+10	
− Base-10	0	2	1	.	1	5	4
	4	3	2	.	0	6	0
	5	8	9	.	0	9	4

Last Borrow → 021.154 < 432.060 → Negative Result

	+10	+10					
-1	-1	-1	+10		-1	+10	
–	0	2	1	.	1	5	4
Base-10	4	3	2	.	0	6	0
	5	8	9	.	0	9	4

The Result is not correct.

Extra Subtraction:

	+10	+10	+10		+10	+10	
-1	-1	-1	-1		-1	-1	+10
1	0	0	0	.	0	0	0
−	5	8	9	.	0	9	4
Base-10	4	1	0	.	9	0	6

$= (021.154)_{10} - (432.060)_{10} = (021.154)_{10} + (1000.000)_{10} - (1000.000)_{10} - (432.060)_{10}]$

$= -(1000.000)_{10} + (589.094)_{10} = -[(1000.000)_{10} - (589.094)_{10}]$

$= -(410.906)_{10}$

SUBTRACTION

Example II, Base-16

PADDING

–	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
				.			

			BORROW		-1	+16	
−	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B=11	0
				.		= 5+16-11 = 10 = A	4

					-1	+16	
−	0	2	A	.	E=14	5	4
Base-16	4	B	F	.	2	B	0
				.	B=11	A	4

		+16					
	-1	-1	+16		-1	+16	
−	0	2	A	.	E	5	4
Base-16	4	B=11	F	.	2	B	0
		6	B	.	B	A	4

		+16					
	-1	-1	+16		-1	+16	
−	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
		6	B	.	B	A	4

	+16	+16					
-1	-1	-1	+16		-1	+16	
−	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
	B=11	6	B	.	B	A	4

	+16	+16					
-1	-1	-1	+16		-1	+16	
−	0	2	A	.	E	5	4
Base-16	4	B	F	.	2	B	0
	B	6	B	.	B	A	4

02A.E54 < 4BF.2B0

Last Borrow → Negative Result

Extra Subtraction:

	+16	+16	+16		+16	+16	
-1	-1	-1	-1		-1	-1	+16
1	0	0	0	.	0	0	0
−	B=11	6	B=11	.	B=11	A=10	4
Base-16	4	9	4	.	4	5	C=12

$$= (2A.E54)_{16} - (4BF.2B0)_{16} = (2A.E54)_{16} + (1000.000)_{16} - (1000.000)_{16} - (4BF.2B0)_{16}$$

$$= -(1000.000)_{16} + (B6B.BA4)_{16} = -[(1000.000)_{16} - (B6B.BA4)_{16}]$$

$$= -(494.45C)_{16}$$

DIVISION

At Home

SIGNED NUMBERS

NEGATIVE | POSITIVE

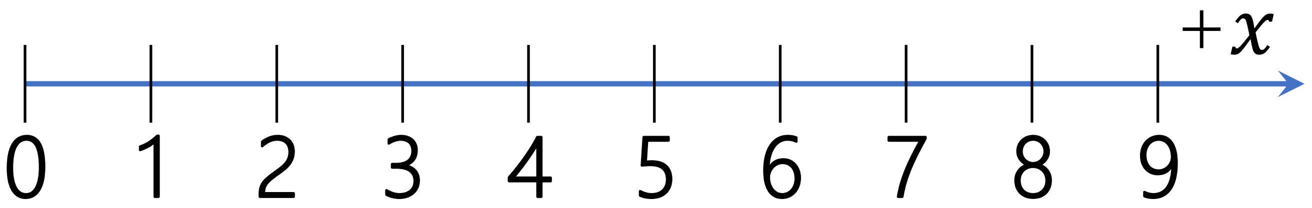
aka Signed Numbers

SIGNED MAGNITUDE
SIGNED COMPLEMENT

SIGNED MAGNITUDE

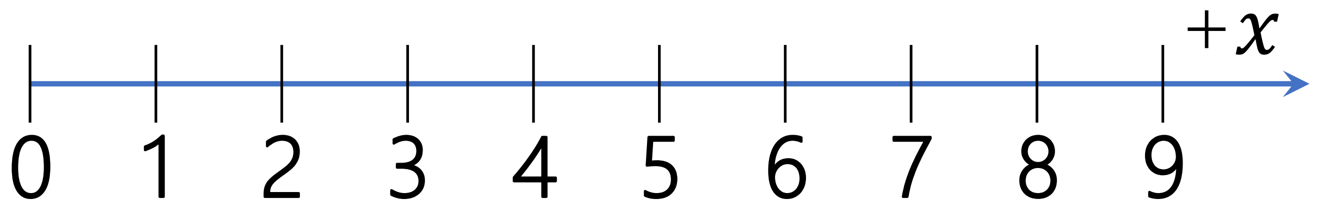
SIGNED COMPLEMENT

10^0
0
1
2
3
4
5
6
7
8
9



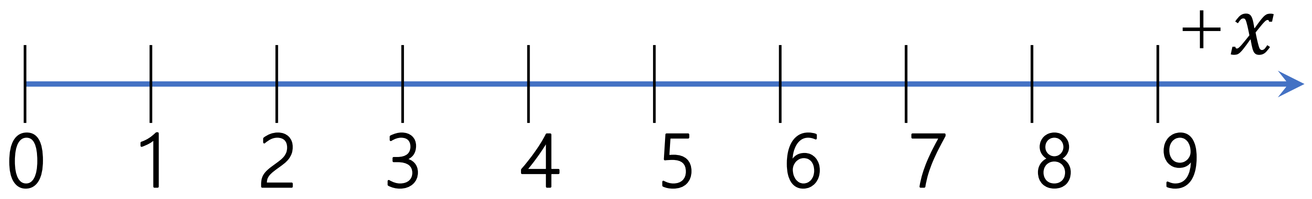
Any idea to represent negative numbers?

10^0
0
1
2
3
4
5
6
7
8
9



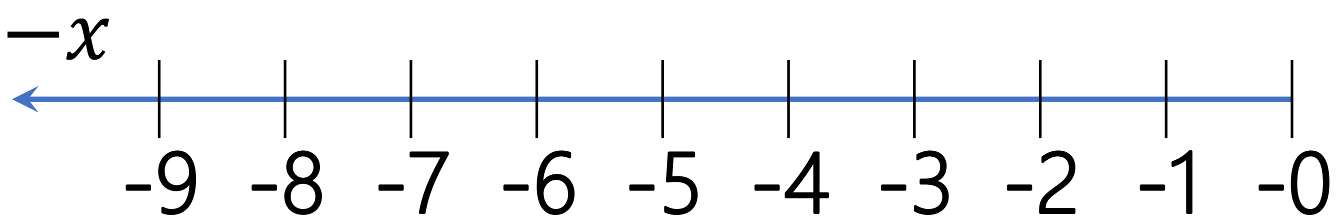
New symbols for each digit?

10^1	10^0
	0
	1
	2
	3
	4
	5
	6
	7
	8
	9



A new position?

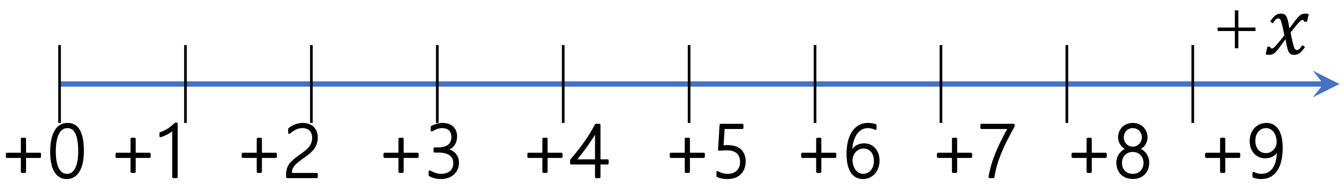
10^1	10^0
–	0
–	1
–	2
–	3
–	4
–	5
–	6
–	7
–	8
–	9



A new position? Yes

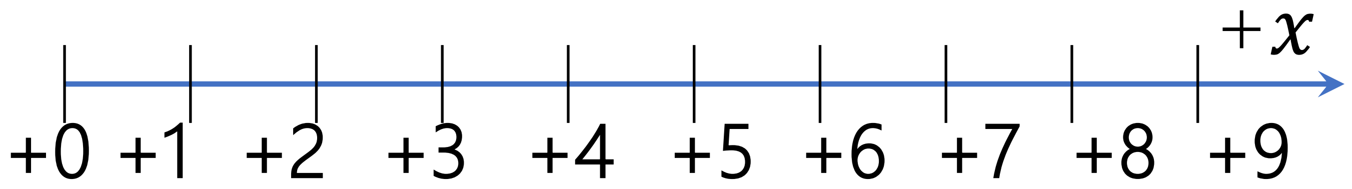
A new symbol: –

10^1	10^0
+	0
+	1
+	2
+	3
+	4
+	5
+	6
+	7
+	8
+	9



A new position? Yes
A new symbol: +

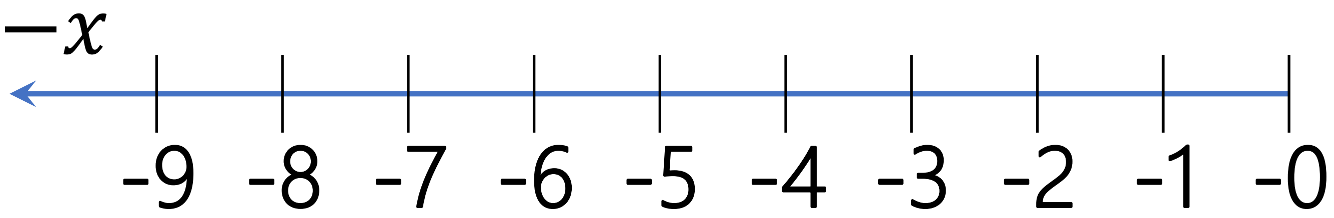
10^1	10^0
0	0
0	1
0	2
0	3
0	4
0	5
0	6
0	7
0	8
0	9



A new position? Yes

~~A new symbol: +~~

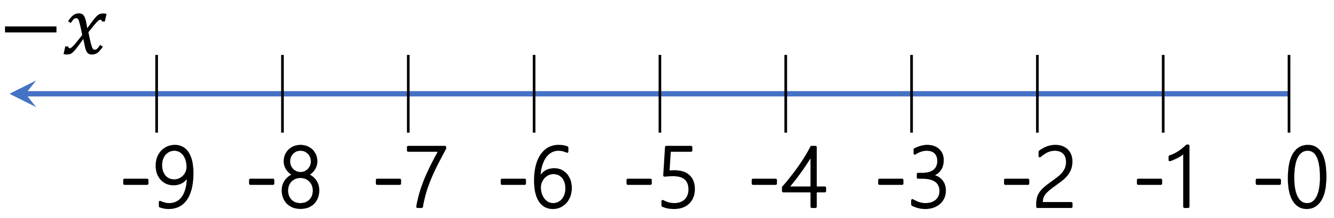
10^1	10^0
1	0
1	1
1	2
1	3
1	4
1	5
1	6
1	7
1	8
1	9



A new position? Yes

~~A new symbol: −~~

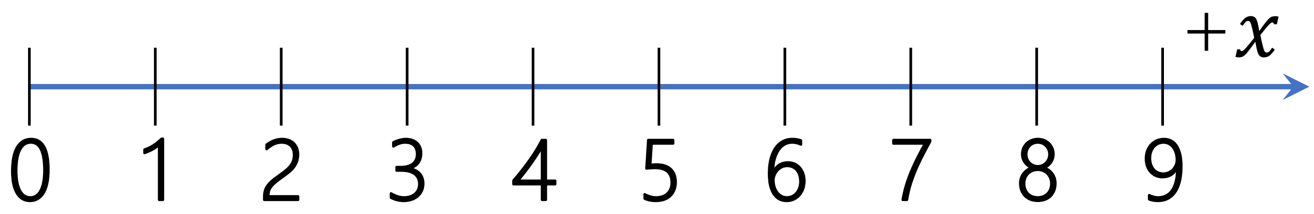
10^1	10^0
Nonzero	0
Nonzero	1
Nonzero	2
Nonzero	3
Nonzero	4
Nonzero	5
Nonzero	6
Nonzero	7
Nonzero	8
Nonzero	9

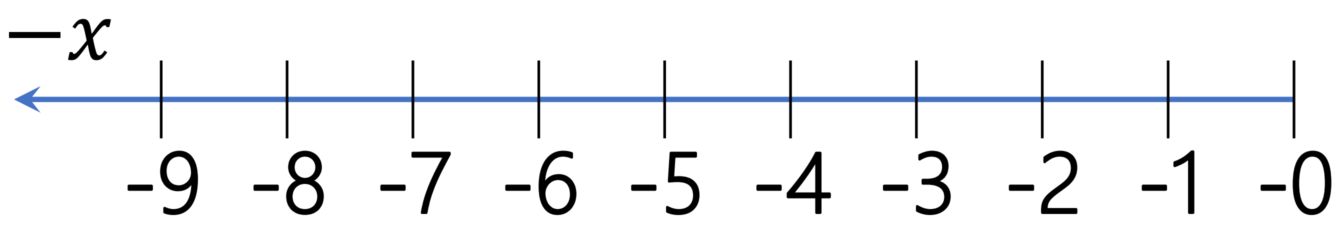


A new position? Yes

~~A new symbol: −~~

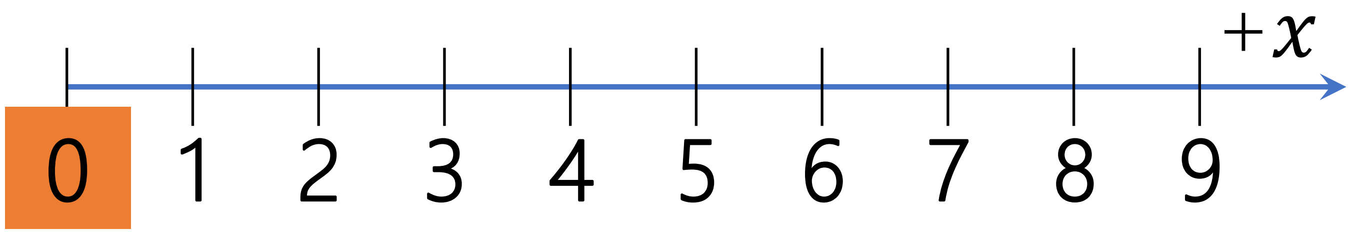
$\cancel{10^1}$	10^0
1,2,…9	0
1,2,…9	1
1,2,…9	2
1,2,…9	3
1,2,…9	4
1,2,…9	5
1,2,…9	6
1,2,…9	7
1,2,…9	8
1,2,…9	9

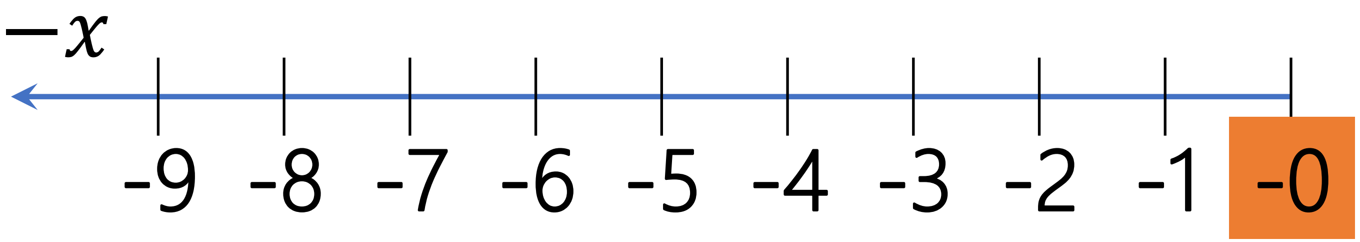




$\cancel{10^1}$	10^0
0	0
0	1
0	2
0	3
0	4
0	5
0	6
0	7
0	8
0	9

$\cancel{10^1}$	10^0
1,2,...9	0
1,2,...9	1
1,2,...9	2
1,2,...9	3
1,2,...9	4
1,2,...9	5
1,2,...9	6
1,2,...9	7
1,2,...9	8
1,2,...9	9





$\cancel{10^1}$	10^0
0	0
0	1
0	2
0	3
0	4
0	5
0	6
0	7
0	8
0	9

SIGNED MAGNITUDE

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

Signed

Magnitude

Give up left most position for sign!

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

$$0 \longrightarrow \text{Max} = r^{n-1} - 1 = \cancel{r^n - 1}$$

$$\text{Min} = -(r^{n-1} - 1) \leftarrow \cancel{0}$$

$\cancel{2^5}$	2^4	2^3	2^2	2^1	2^0
0	0	0	0	1	1
1	0	0	1	0	1

Interpretation	Base-10	
?	?	?
?	?	?

$\cancel{7^5}$	7^4	7^3	7^2	7^1	7^0
0	0	0	0	5	6
1	0	0	2	0	5
4	0	0	2	0	5
6	0	0	2	0	5

Interpretation	Base-10			
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?

$\cancel{2^5}$	2^4	2^3	2^2	2^1	2^0
0	0	0	0	1	1
1	0	0	1	0	1

Interpretation	Base-10	
$+(3)_{10}$	0	3
$-(5)_{10}$	1,2,...9	5

$\cancel{7^5}$	7^4	7^3	7^2	7^1	7^0
0	0	0	0	5	6
1	0	0	2	0	5
4	0	0	2	0	5
6	0	0	2	0	5

Interpretation	Base-10			
$+(41)_{10}$	0	0	4	1
$-(103)_{10}$	1,2,...9	1	0	3
$-(103)_{10}$	1,2,...9	1	0	3
$-(103)_{10}$	1,2,...9	1	0	3

There Is No Symbol For '-' , '+'
Just Our Interpretation!

SIGNED MAGNITUDE ARITHMETIC

0	X	$+$	0	Y	$=$	0	$X+Y$
0	X	$+$	1	Y	$=$	$X<Y$ (if borrow)	$X-Y$ (if borrow, apply it)
1	X	$+$	1	Y	$=$	1	$-(X+Y) = X+Y$

0	X	$-$	0	Y	$=$	$X<Y$ (if borrow)	$X-Y$ (if borrow, apply it)
0	X	$-$	1	Y	$=$	0	$X+Y$
1	X	$-$	1	Y	$=$	$X>Y$ (if borrow)	$-X+Y=Y-X$ (if borrow, apply it)

SIGNED MAGNITUDE

Example I, Addition, Base-16

+	1	2	A	.	E	5	4
Base-16	1	B	F	.	2	B	
				.			

+	1	2	A	.	E	5	4
Base-16	1	B	F	.	2	B	
				.			

PADDING

+	1	2	A	.	E	5	4
Base-16	1	B	F	.	2	B	0
				.			

SIGNED: $(-X)+(-Y)=-(X+Y)$

+	1	2	A	.	E	5	4
Base-16	1	B	F	.	2	B	0
	1			.			

		1	1		1		
+	1	2	$A_{=10}$	.	$E_{=14}$	5	4
Base-16	1	B	$F_{=15}$	.	2	$B_{=11}$	0
	1	$E_{=14}$	$A_{=10}$	.	1	0	4

SIGNED MAGNITUDE EXAMPLE II

+ Base-16							
	0	2	A	.	E	5	4
	1	B	F	.	2	B	
				.			

+ Base-16	0	2	A	.	E	5	4
	1	B	F	.	2	B	
				.			

PADDING

+ Base-16	0	2	A	.	E	5	4
	1	B	F	.	2	B	0
				.			

SIGNED: $(+X)+(-Y)=?(X-Y)$

−	0	2	A	.	E	5	4
Base-16	1	B	F	.	2	B	0
	?			.			

	−1 →	+16					
		−1	+16		−1	+16	
−	0	2	A=10	.	E=14	5	4
Base-16	1	B=11	F=15	.	2	B=11	0
	1	6	B=11	.	B=11	A=10	4

2A.E54 < BF.2B0

Last Borrow → Negative Result

		+16	+16		+16	+16	
	−1	−1	−1		−1	−1	+16
−	1	0	0	.	0	0	0
Base-16		6	B=11	.	B=11	A=10	4
		9	4	.	4	5	C=12

$$= (2A.E54)_{16} - (BF.2B0)_{16} = (2A.E54)_{16} + (100.000)_{16} - (100.000)_{16} - (BF.2B0)_{16}$$

$$= -(100.000)_{16} + (6B.BA4)_{16} = -[(100.000)_{16} - (6B.BA4)_{16}]$$

$$= -(94.45C)_{16}$$

	−1 →	+16					
		−1	+16		−1	+16	
−	0	2	A=10	.	E=14	5	4
Base-16	1	B=11	F=15	.	2	B=11	0
	1	6	B=11	.	B=11	A=10	4

2A.E54 < BF.2B0

Last Borrow → Negative Result

1	9	4	.	4	5	C

SIGNED MAGNITUDE
OVERFLOW

Overflow!

0	X	+	0	Y	=	$0 \rightarrow 1$	$X+Y$
0	X	+	1	Y	=	$X<Y$	$X-Y$
1	X	+	1	Y	=	$1 \rightarrow 1$	$X+Y$

0	X	$-$	0	Y	=	$X<Y$	$X-Y$
0	X	$-$	1	Y	=	$0 \rightarrow 1$	$X+Y$
1	X	$-$	1	Y	=	$X>Y$	$-X+Y=Y-X$

− Base-2	$+(27)_{10}$	0	1	1	0	1	1
		1	0	1	1	1	0

$-$ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$-(14)_{10}$	1	0	1	1	1	0

$+X - (-Y)$

$-$ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$-(14)_{10}$	1	0	1	1	1	0

$$+X - (-Y) = +(X + Y)$$

+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0

$$+X - (-Y) = +(X + Y)$$

+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		0					

$$+X - (-Y) = +(X + Y)$$

		1	1	1	1		
+	$+(27)_{10}$	0	1	1	0	1	1
Base-2	$+(14)_{10}$	0	0	1	1	1	0
		0	0	1	0	0	1

$$+X - (-Y) = +(X + Y)$$

		1	1	1	1		
+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		1	0	1	0	0	1
	If you consider the last carry → $-(9)_{10}$ → Negative!						

$$+X - (-Y) = +(X+Y)$$

		1	1	1	1		
+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		0	0	1	0	0	1
	If you ignore it → $+(9)_{10}$ → Result is not correct!						

$$+X - (-Y) = +(X+Y)$$

		1	1	1	1		
+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
	Overflow: The result is not reliable!						

PRACTICE

1. Base (r)
2. First Number (X)
3. Second Number (Y)
4. Signed or Unsigned?
    1. If Signed, What Signed System?
5. Operation?
6. Overflow?

PRACTICE (chatgpt?)

1. Base (r=7)
2. First Number (X=023.31)
3. Second Number (Y=121.46)
4. Signed or Unsigned? Yes
    1. If Signed, What Signed System? Signed Magnitude
5. Operation? Subtraction
6. Overflow?

WHY NOT SIGNED MAGNITUDE

Give up left most position for sign! What are the wastes?

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

$$+0 \rightarrow \text{Max} = r^{n-1} - 1 = \cancel{r^n - 1}$$

$$\text{Min} = -(r^{n-1} - 1) \leftarrow -0$$

SIGNED MAGNITUDE

SIGNED COMPLEMENT